
Social OMNI-EPIC: Learning-Progress-Driven Curriculum Generation for Social Interaction

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Abstract

We introduce Social OMNI-EPIC, an open-ended curriculum generation framework that adapts learning-progress-driven environment design to social interaction. Starting from a seed archive, the system continuously generates, evaluates, and expands social scenarios toward a learner’s capability frontier using intrinsic signals of learning progress and interestingness. Across 90 generated scenarios, Social OMNI-EPIC shifts the distribution from predominantly easy scenarios toward frontier-level challenges and becomes increasingly calibrated over time.

1 Introduction

Social intelligence, the ability to navigate interpersonal interaction through negotiation, persuasion, empathy, and cooperation, is not innate but develops through experience. Developmental accounts from Vygotsky’s sociocultural theory [Vygotsky, 1978] to Tomasello’s comparative work [Tomasello, 2019] converge on a common theme: social competence develops progressively, through encounters of appropriate difficulty at the frontier of the learner’s current ability.

This mirrors a principle from machine learning, that a training signal is useful only when it is neither too easy nor too hard, formalized in curiosity-driven research as *learning progress* and used as an intrinsic reward to grow developmental curricula [Oudeyer et al., 2007]. OMNI-EPIC [Faldor et al., 2025] operationalizes this for open-ended environment generation, having a foundation model continuously propose environments that are both learnable and interestingly different from those already seen. This machinery has not reached the social domain, where it is arguably most needed. Large language models are increasingly used as social agents [Park et al., 2023], but existing platforms evaluate them on fixed scenario sets [Zhou et al., 2024b] that cannot reveal where a given model still struggles, and naive synthetic expansion is unreliable because preference-tuned models drift toward agreeableness and uniformity [Sharma et al., 2024, Anthis et al., 2025]. The social training curriculum therefore remains static and hand-curated.

We propose Social OMNI-EPIC, which adapts learning-progress-driven generation to dyadic social interaction. The system continuously generates scenarios that are both learnable by a frozen learner and interestingly distinct from those already seen, locating difficulty automatically at the learner’s frontier. A central difference from the original setting is that social competence is graded rather than a binary success, providing a continuous gradient for the learning-progress signal to climb. The outcome is an ever-expanding scenario bank calibrated to a model’s social frontier, producing an experience profile that improves the model without weight updates.

2 Related Work

Open-ended and unsupervised environment design study how to produce the training distribution itself. Methods in unsupervised environment design adapt environment parameters to a learner,

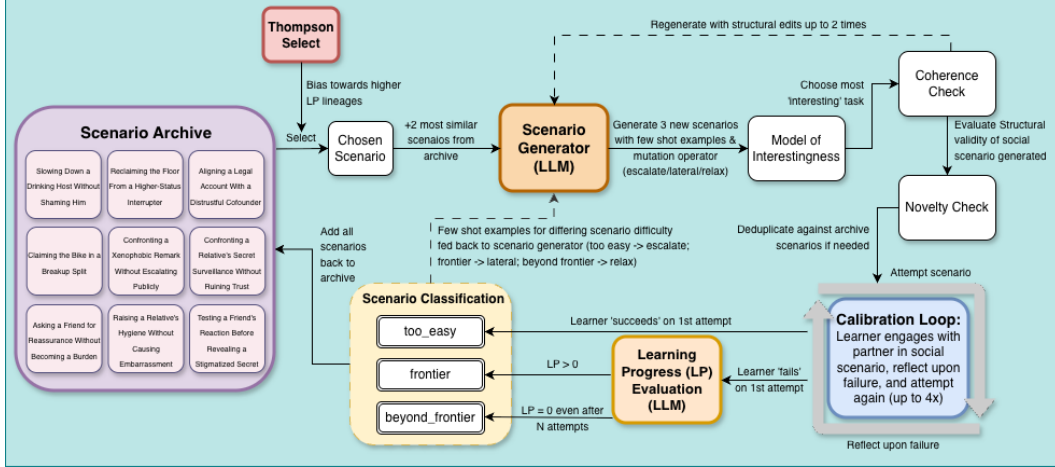


Figure 1: **Social OMNI-EPIC Curriculum Loop.** The system maintains a scenario archive and repeatedly expands it toward the frontier of a frozen learner’s social competence. Thompson sampling, biased toward lineages with high learning progress, selects a scenario, which the generator mutates into new candidates using the operator implied by the parent’s class (escalate, lateral, or relax). The model of interestingness keeps the most promising candidate, which then passes a coherence check and a novelty check against the archive before the learner engages the partner in the calibration loop, reflecting on failure and reattempting up to four times. A learning-progress evaluation then labels each scenario too-easy, frontier, or beyond-frontier, and all scenarios are written back to the archive, where the labels feed both the next round of selection and the operator choice

selecting or generating levels by criteria such as regret [Dennis et al., 2020, Jiang et al., 2021, Parker-Holder et al., 2022]. POET co-evolves environments together with the agents that solve them, so that newly generated challenges remain within reach as agents improve [Wang et al., 2019, 2020]. While environment-centered formulations often locate difficulty in environment parameters alone, multi-agent variants place part of the difficulty in a co-player whose behavior shapes the challenge [Samvelyan et al., 2023], which is relevant whenever the source of difficulty in a scenario is another agent. A limitation common to these methods is their dependence on a predefined space of environment parameters, which bounds the situations that can be generated [Faldor et al., 2025].

OMNI introduces a foundation model as a model of interestingness, used to choose, among the tasks on which a learner can make progress, those that are also worthwhile and sufficiently novel [Zhang et al., 2024]. It keeps the two judgements separate: a learning-progress curriculum determines which tasks lie at an appropriate level of difficulty, and the model of interestingness filters out tasks that are learnable but not worth learning. OMNI-EPIC extends this approach by having a foundation model generate environments directly as code, which removes the dependence on a predefined parameter space and allows the curriculum to propose, in principle, any situation expressible as a program [Faldor et al., 2025]. In OMNI-EPIC, a generated task is admitted to the archive once a learner is judged to have completed it, a determination made by a success detector that returns a binary verdict. Our work builds on this open-ended, foundation-model-driven generation but targets a domain in which the relevant measure of competence is graded rather than binary, which changes what the curriculum’s selection signal can be. We describe that adaptation next.

3 Method

Social OMNI-EPIC adapts the open-ended curriculum generation of OMNI-EPIC [Faldor et al., 2025] to social interaction. The system holds a single frozen learner and grows an archive of social scenarios whose difficulty tracks that learner’s competence frontier. We define ‘frontier’ as a situation when the learner cannot solve a scenario from its trained priors alone but can improve once augmented with the right knowledge in context. A scenario pairs the learner with a partner agent in a goal-directed social situation. The learner’s weights are never updated, and improvements within a single scenario come from the learner reflecting on its own prior attempts (Section 3.2). The contribution of this work is

63 the loop itself: an open-ended and learning-progress-driven source of social scenarios calibrated to a
64 given learner (Figure 1).

65 3.1 Social Scenarios & Constructed Resistance

66 Because the curriculum generates and expands scenarios without a human author in the loop, it cannot
67 lean on human judgement to keep each scenario coherent, solvable, and genuinely difficult. A fixed,
68 theory-grounded schema plays that role instead, constraining the generator the way a careful author
69 would so that what it produces is well-posed by construction. We build on SOTOPIA’s scenario
70 and agent-profile schema [Zhou et al., 2024b] and add two structured layers, one on each side of
71 the interaction. The agents still interact in a SOTOPIA-native setting, since the structured goal is
72 rendered back into an ordinary SOTOPIA goal string for play; the structure is a generation-time
73 scaffold underneath, not a change to what the agents see.

74 The first layer is the learner’s goal, stated as three linked elements. An *outcome* is what the learner is
75 trying to bring about; a *constraint* is a relational or normative boundary the outcome must not violate;
76 and a *shortcut* is an efficient move that secures the outcome while breaking the constraint. The split
77 between outcome and constraint follows the goals-plans-action distinction between the primary goal
78 that drives an influence attempt and the secondary, often relational or identity-protecting, goals that
79 bound how it may be pursued [Dillard et al., 1989], and the constraint is most often a face threat to
80 be avoided in the sense of Brown and Levinson [Brown, 1987]. The shortcut is a design field that
81 fixes the naive failure mode the generator must build in, and a learner that takes it is penalized only
82 indirectly through the constraint check it thereby fails. Stating the goal as an explicit triple lets the
83 generator and the coherence check (Section 3.3) reason over named parts, so refining a scenario’s
84 ‘interestingness’ is a targeted edit rather than an unguided rewrite.

85 The two sides are not symmetric. The learner is the agent whose competence the curriculum tracks
86 and the only one given the goal triple; the partner instead carries a natural-language goal and the
87 resistance structure described next.

88 Unlike the learner’s structured goal triplet, the partner holds a different, hidden resistance structure
89 we call the **partner key**. This key is never shown to the learner and absent from the shared scenario
90 text, is supplied only to the partner, and is used both to drive the partner’s behavior and to check at
91 the end whether the learner met its conditions (Section 3.2). A key pairs a resistance mechanism,
92 drawn from a small but expandable library of theory-grounded mechanisms, with four fields that
93 instantiate it for the scenario. See Appendix A for more details.

94 We build scenarios around a partner key for three reasons. First, it is what keeps difficulty from
95 dissolving into mutual politeness. Preference-tuned models default to agreeableness [Sharma et al.,
96 2024], and in the realistic, information-asymmetric setting they often reach an apparent agreement
97 without genuinely engaging with the social tension [Zhou et al., 2024a], so a scenario whose resistance
98 rests on a stated personality description tends to resolve trivially. The key replaces a personality label
99 with explicit rules the resistance must follow. A separate field records the partner’s default position,
100 the offer it makes without skilled engagement, and we require that default to fall short of the learner’s
101 outcome so a real gap remains. Second, requiring a key forces the generator to first lay out a rigorous
102 schema for the social puzzle, including at least one deliberately non-exhaustive path that satisfies
103 the movement conditions without tripping the hardening triggers, which guarantees a scenario is
104 hard rather than impossible. Third, because the partner is given the same resistance disposition on
105 every retry, it behaves consistently across the learner’s attempts, so a change in measured competence
106 reflects the learner improving rather than the partner drifting. We checked this directly: across five
107 hand-authored scenarios replayed three times each, the partner held to its key on 14 of 15 runs.

108 3.2 Measuring Competence & Learning Progress

109 A scenario is played out in the calibration loop. The learner engages the partner and, when an attempt
110 does not succeed, reflects on the failure and tries again, for up to four attempts in total [Shinn et al.,
111 2023]. Each attempt is scored on the goal and relationship dimensions of SOTOPIA-EVAL [Zhou
112 et al., 2024b], and an attempt only counts as success if it meets our threshold $GOAL \geq 7$ AND REL
113 ≥ 0 adapted from SOTOPIA-EVAL [Zhou et al., 2024b]. Within the calibration loop the learner is
114 calibrating its own approach to a single scenario, and how much it can improve across attempts is the
115 learning-progress signal that the wider curriculum loop reads.

Learning progress is scored by a separate judge, drawn from a different model family than the learner and blind to the partner’s resistance structure. The judge takes the first attempt as an anchor and compares it against each later attempt. Every comparison is run in both presentation orders, and when the two orders disagree the comparison collapses to no difference, which removes position bias. The judge scores holistically: a later attempt counts as better only if it makes genuine progress on the goal while at least preserving the relationship, so neither extracting the outcome at the cost of the partner’s trust nor improving only in warmth or politeness without goal progress counts as improvement. Learning progress is the fraction of comparisons in which the later attempt is judged better, and lies between 0 and 1.

These two signals classify the scenario. A scenario whose first attempt succeeds is too-easy. Otherwise it is frontier when learning progress is positive and beyond-frontier when learning progress is zero across the four attempts. This notion of learning progress is a within-episode, reflection-driven quantity, and is distinct from the classical learning-progress signal defined as a derivative of competence accumulated over weight updates.

3.3 Generating New Scenarios

New scenarios are produced by expanding an existing one. A single operator governs each expansion and is fixed by the parent’s class: a ‘too-easy’ parent is escalated, a ‘beyond-frontier’ parent is relaxed, and a ‘frontier’ parent is explored laterally. The operator names only a difficulty direction relative to the parent and does not edit the parent’s text and does not predict where the child will land. As such, every child scenario is generated with a completely fresh surface with new characters, setting, and stakes that is still within the parent’s structural family. Where the scenario actually lands is settled by the learning-progress measurement of Section 3.2 rather than asserted at generation.

Generating one child runs a short pipeline. The generator is prompted with the parent, the chosen direction, and a few similar scenarios retrieved from the archive, and is asked for situations that are interesting, learnable, yet difficult by default, where difficult means a model that simply defers and agrees should fail to reach the learner’s outcome. It returns several candidates, and a model of interestingness (MOI) ranks them on the axis of social worth: whether the tension is one a thoughtful person would recognize as a real human situation rather than a contrived puzzle. The candidate rated highest by the MOI is kept.

The kept candidate then passes a coherence check, which is where the partner-key invariants are enforced. Beyond verifying that the scenario is internally consistent and playable, it checks that the movement conditions do not appear in any text the learner can see, that the shortcut mechanically trips at least one hardening trigger rather than merely being judged wrong after the fact, that satisfying the partner’s conditions costs the learner something real yet still leaves a reachable version of the outcome, and that the scenario cannot be won by simply accepting whatever the partner first offers. A candidate that fails is regenerated with targeted edits up to twice. The full check list is given in Appendix B.

3.4 Curriculum Control Flow

Social OMNI-EPIC repeatedly selects a parent from the archive, generates a difficulty-shifted variant, tests whether the frozen learner can make progress on it, and feeds the outcome back to guide future selection. The result is a self-curating curriculum that concentrates generation on scenarios that are hard but learnable. The full procedure is given in Appendix C.

Scenarios as Beta arms Each archived scenario is a bandit arm whose latent payoff is how useful a learning target its children are. We model that belief as $\text{Beta}(a_0 + a_v, b_0 + b_v)$, with prior (a_0, b_0) and votes (a_v, b_v) accumulated from the scenario’s children. A generated child inherits a warm-started prior from its parent, capped so deep lineages do not compound unwarranted confidence.

Selection Each round we draw one independent sample from every arm’s posterior and take the B highest as parents (Thompson sampling), which exploits arms believed productive while still revisiting uncertain ones, with no tuned exploration constant.

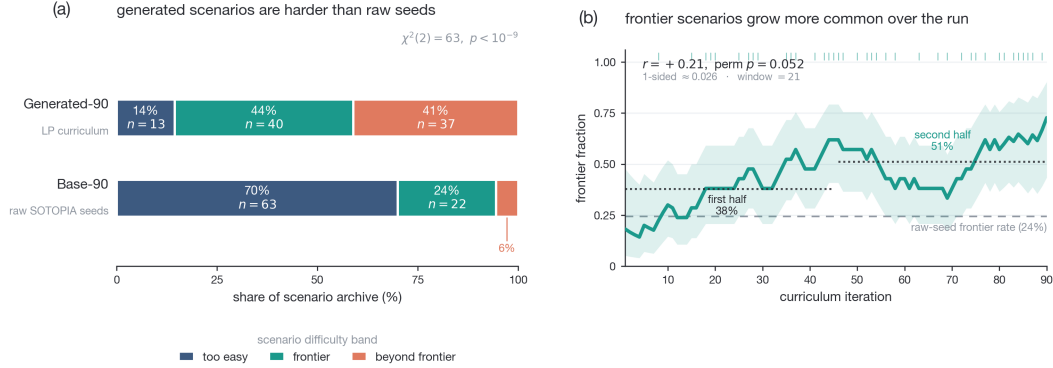


Figure 2: Curriculum calibration toward the learning frontier. (a) Difficulty distribution of 90 raw Sotopia seed scenarios versus 90 scenarios generated by Social OMNI-EPIC. (b) Fraction of frontier scenarios over the course of curriculum generation.

165 **Generation, classification, and feedback** We treat each of a child’s pairwise comparisons as a
 166 Bernoulli trial on the parent: comparisons in which a later attempt improved credit the success side of
 167 its posterior, and the rest credit the failure side. A frontier child therefore contributes its improved and
 168 not-improved counts; a beyond-frontier child, showing no improvement, contributes only failures;
 169 and a too-easy child, solved on the first attempt and never reaching the comparison stage, contributes
 170 a fixed block of failure mass. Generation failures incur a small penalty and are discarded. An arm’s
 171 posterior mean is thus the improvement rate of its children, so selection drifts toward parents that
 172 produce learnable-but-unsolved scenarios while the band-conditioned operators steer difficulty to the
 173 frontier.

174 4 Results & Discussion

175 Our evaluation addresses two questions: (1) whether the curriculum can calibrate and self-organize
 176 toward the agent’s learning frontier, and (2) whether an experience bank built from curriculum-
 177 generated scenarios improves a frozen agent on held-out social tasks through in-context learning.

178 For the first question, We generated 90 new scenarios using the full Social OMNI-EPIC loop and
 179 compared their difficulty distribution against the 90 raw SOTOPIA seed scenarios. Figure 2 (a) shows
 180 a striking shift in difficulty distribution. The raw SOTOPIA seeds are dominated by scenarios that are
 181 too easy for the agent (70%), with only 24% at the frontier and 6% beyond it. In contrast, the generated
 182 scenarios are concentrated at and beyond the frontier: 44% frontier and 41% beyond frontier, with
 183 only 14% too easy. This difference is highly significant ($\chi^2(2) = 63, p < 10^{-9}$), confirming that the
 184 learning-progress signal substantially reshapes the difficulty distribution of generated scenarios.

185 Figure 2 (b) further shows that this calibration improves over the course of the run. The fraction of
 186 frontier scenarios increases from an average of 38% during the first half of the run to 51% in the
 187 second half. This upward trend suggests that the generation loop does not simply produce harder
 188 scenarios on average; rather, it progressively tunes its output toward the agent’s evolving capability
 189 boundary as the archive grows and the learning-progress signal accumulates.

190 To answer the second question, we evaluated model performance on 150 held-out scenarios sam-
 191 pled uniformly from the SOTOPIA- π extended scenario set [Wang et al., 2024]. The learner
 192 and its conversational partner are both gpt-5-mini; episodes are scored by a cross-lab judge,
 193 gemini-3-flash-preview, to avoid same-model evaluation bias. Conditions differ only in the
 194 experience available at inference time, using an in-context learning method called ExpeL [Zhao et al.,
 195 2024] which 1) gathers reflective trajectories on the training scenarios, 2) distills them into a compact
 196 set of natural-language rules, and 3) at test time retrieves the most relevant rules and trajectories into
 197 the learner’s context (See Appendix D)

198 We compare four conditions: **Vanilla**, the frozen model with no retrieved experience; **Random-90**,
 199 an ExpeL bank built from 90 randomly sampled SOTOPIA- π scenarios(disjoint from seeds and test);

	Avg. GOAL	Avg. REL	Success	Δ Succ.
<i>All scenarios</i>				
Vanilla	8.75	2.67	83.3%	—
Random-90	8.75	2.74	85.3%	+2.0
ExpeL-90	8.66	2.81	81.3%	−2.0
SOE (ours)	8.48	2.72	80.5%	−2.8
<i>Vanilla-failed subset ($n = 25$)</i>				
Vanilla	5.40	1.84	0.0%	—
Random-90	6.40	2.36	40.0%	+40.0
ExpeL-90	6.28	2.12	32.0%	+32.0
SOE (ours)	5.84	2.20	40.0%	+40.0

Table 1: Held-out evaluation. **All**: all held-out scenarios. **Vanilla-failed**: the $n = 25$ scenarios where Vanilla fails ($\text{GOAL} < 7$ or $\text{REL} < 0$). $\text{GOAL} \in [0, 10]$, $\text{REL} \in [-5, 5]$; Success is the fraction with $\text{GOAL} \geq 7$ and $\text{REL} \geq 0$. Δ is the change in Success relative to Vanilla (in points). Best per block in bold.

200 **ExpeL-90**, a bank built from the 90 SOTOPIA seed scenarios; and **SOE (ours)**, a bank built from the
201 curriculum our loop generates from those same 90 seeds.

202 What differs across conditions is only the set of scenarios that fills stages (i)–(ii); the learner, retriever,
203 and prompt format are held fixed.

204 Table 1 reports performance on the held-out set. Across all scenarios, the four conditions perform
205 similarly; the empty-bank Vanilla baseline already succeeds on 83.3% of scenarios, leaving little
206 headroom for retrieved experience to improve aggregate performance.

207 The differentiating signal emerges on the 25 scenarios that Vanilla fails to solve. On this subset, every
208 ICL condition recovers a substantial fraction of failures: Random-90 and SOE both increase success
209 from 0% to 40.0%, while ExpeL-90 reaches 32.0%. These gains are accompanied by corresponding
210 improvements in average GOAL ($5.40 \rightarrow 5.84\text{--}6.40$) and REL ($1.84 \rightarrow 2.12\text{--}2.36$). Retrieved
211 experience therefore provides clear benefits precisely where the base model struggles.

212 However, the curriculum-generated bank does not clearly separate from the non-curriculum banks.
213 SOE matches Random-90 in recovered success rate but trails it slightly in average GOAL and REL.
214 We interpret these results as an evaluation-ceiling effect rather than evidence against the curriculum.
215 The held-out set is dominated by 125 scenarios the frozen model solves on the first pass, leaving
216 only 25 genuinely hard cases, which are too few to resolve fine-grained differences between ICL
217 banks that all recover roughly a third to 40% of them. Critically, Random-90’s experience bank is
218 drawn from SOTOPIA- π extended, the very source the held-out scenarios are sampled from. On the
219 other hand, our curriculum deliberately generates scenarios beyond it. The curriculum’s intended
220 advantage, which are harder, compositional situations past the seed distribution, thus has little to
221 surface when the test set itself is similarly-sourced and largely solvable.

222 The ideal test would draw hard scenarios from a source independent of both our pipeline and
223 SOTOPIA, as generating them with our own loop would confound the curriculum and the evaluation.
224 We therefore take the present result as establishing that inference-time retrieval reliably recovers
225 base-model failures, and leave a harder, independently sourced benchmark to future work.

226 5 Conclusion

227 Social OMNI-EPIC demonstrates that learning progress can serve as an intrinsic objective for
228 open-ended social curriculum generation, producing scenario archives that increasingly concentrate
229 around a learner’s capability frontier. While curriculum-generated experience did not outperform
230 non-curriculum baselines on an easy held-out benchmark, the framework provides a foundation for
231 studying adaptive social learning in more challenging and open-ended settings.

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302 A Scenario Schema and Partner Key

303 A scenario extends SOTOPIA’s scenario and agent-profile schema with two structured layers. The
 304 learner (agent 0) carries the goal triple of Section 3.1: an *outcome* (the instrumental ask), a *constraint*
 305 (the relational or face cost the blunt path would incur), and a *shortcut* (the tempting move that wins
 306 the outcome but breaks the constraint). The shortcut drives generation and the naive failure mode
 307 but is not itself a rubric check as it is penalized only through the constraint check it causes to fail.
 308 The partner (agent 1) carries a natural-language goal and a hidden *partner key*, and both agents
 309 additionally carry a persona profile (name, age, occupation, Big-Five traits, values, decision style,
 310 and a private secret) rendered into the SOTOPIA-native interaction.

311 The partner key has five fields: a *mechanism* drawn from the library below, and four fields that
 312 instantiate it for the scenario, namely 1. *movement conditions* (one to three concrete conditions under
 313 which the partner genuinely shifts), 2. *hardening triggers* (one to three learner moves that make
 314 it resist more firmly), 3. a *surface misdirection* (the stated public stance, which may appear in the
 315 shared text), and 4. a *cost coupling* (what satisfying the movement conditions costs the learner’s
 316 own goal). Two further cooperative-alignment fields guard against trivial scenarios: a *competing*
 317 *interest* that full accommodation would forfeit, and a *partner default position* that must fall short of
 318 the learner’s outcome.

319 B Coherence Checks

320 Before a generated candidate is played, a coherence check flags internal inconsistencies and enforces
 321 the partner-key invariants. A free non-model pre-check first rejects candidates in which a movement
 322 condition appears near-verbatim in the shared text, or in which a movement condition or hardening
 323 trigger requires an off-channel artifact (a written guarantee, signed document, contract, or emailed
 324 confirmation) rather than something satisfiable in spoken turns. A candidate that clears the pre-check
 325 is then judged on:

- 326 1. **Relationship consistency:** the relationship label matches the relationship background.

Table 2: Theory-grounded resistance mechanisms. The mechanism field of a partner key takes one of these values; the other four fields instantiate it for the scenario.

Mechanism	What moves the partner
Reactance	Pressure and ultimatums harden resistance; restoring autonomy and choice opens it ([Brehm and Brehm, 2013])
Face needs	Movement requires a face-saving account or an identity-preserving exit (Brown [1987])
Validation before change	The partner cannot consider change until it feels understood; premature problem-solving stalls it (motivational interviewing [Iarussi et al., 2013])
Procedural voice	The partner accepts worse outcomes given real voice in the process and rejects imposed ones
Reciprocity and disclosure	Movement is unlocked by the learner’s costly first move: a concession, self-disclosure, or accepted risk ([Carpenter and Greene, 2015])

2. **Goal distinctness:** the two agents pursue genuinely different objectives (the partner having no goal triple is expected).
3. **Profile-goal plausibility:** each agent’s occupation, personality, and decision style make its goal believable.
4. **Scenario-interaction match:** the scenario text describes the stated interaction type.
5. **Key consistency:** the movement conditions are concrete and behaviorally checkable, and each instantiates the declared mechanism.
6. **Key-narrative separation:** the movement conditions appear in no text the learner can see, the hardening-trigger consequences are not revealed in the shared text, and the partner’s goal states what it cares about without naming the conditions that would move it.
7. **Cost coupling:** satisfying the movement conditions costs the learner something stated in its own goal, yet leaves a reachable version of the outcome rather than making it strictly unreachable.
8. **Shortcut-trigger coupling:** the learner’s shortcut trips at least one hardening trigger, so it fails mechanically rather than only by a judge’s verdict.
9. **Cooperative alignment:** the learner cannot win by freely accommodating the partner; either a competing interest is forfeited under full accommodation, or the partner’s default offer falls short of the learner’s outcome.

C Curriculum Control Law

Each scenario s stores a Beta posterior with prior $(a_0(s), b_0(s))$ and accumulated learning-progress votes $(a_v(s), b_v(s))$. $\mathcal{A}$ is the archive, X the batch of B parents selected per round, and op a parent’s mutation operator. PROPOSE-RUN-CLASSIFY generates, runs, and classifies a child, returning its state, the child c , and the $v_\uparrow$ of v_{tot} order-swapped pairwise judge votes that found a later attempt improved over the first. Frontier-judged seeds receive a mildly optimistic prior; all others start uninformative.

D Implementation and Evaluation Details

Models. The learner and partner are both gpt-5-mini. Learning progress is scored by gemini-3-flash-preview, a model from a different family, to avoid same-model evaluation bias.

Hyperparameters. The calibration loop runs up to $K = 4$ attempts. A too-easy child, solved on the first attempt and never reaching the comparison stage, contributes a fixed block of $= 6$ failure votes to its parent’s posterior; this equals the maximum number of order-swapped comparisons a child can otherwise produce ($K - 1 = 3$ later attempts, each judged in both presentation orders). Seeds previously judged frontier-level receive a mildly optimistic Beta(2, 1) prior and all others an

Algorithm 1: Curriculum control law (module internals in Fig. 1)

Input: seeds S_0 ; batch size B ; target count N ; too-easy mass κ ; warm-start cap μ

```
1  $\mathcal{A} \leftarrow S_0$ ;  $g \leftarrow 0$ ; // seeds form the archive;  $g$  counts completed children
2 while  $g < N$  do
3    $X \leftarrow$  the  $B$  tasks  $x$  maximizing independent draws
    $\theta_x \sim \text{Beta}(a_0(x)+a_v(x), b_0(x)+b_v(x))$ ; // Thompson sampling, with
   replacement
4   for  $x \in X$  do in parallel
5      $op \leftarrow$  ESCALATE/RELAX/LATERAL if  $x$  is TOO-EASY / BEYOND-FRONTIER /
     FRONTIER;
6      $R_x \leftarrow \text{PROPOSE-RUN-CLASSIFY}(x, op)$ ; // returns state, child  $c$ , votes
      $(v_\uparrow, v_{\text{tot}})$ 
   // All archive and posterior writes happen here, single-threaded
7   foreach  $x \in X$  do
8     if  $R_x$  failed then
9        $b_v(x) += 1$ ; // small penalty; child not archived
10    else
11       $(a_0, b_0)(c) \leftarrow \text{WARMSTART}(x, \mu)$ ; add  $c$  to  $\mathcal{A}$ ;  $g += 1$ ;
12       $(a_v, b_v)(x) += \begin{cases} (0, \kappa) & \text{if TOO-EASY} \\ (v_\uparrow, v_{\text{tot}} - v_\uparrow) & \text{if FRONTIER} \\ (0, v_{\text{tot}}) & \text{if BEYOND-FRONTIER} \end{cases}$ ;
```

361 uninformative Beta(1, 1). Warm-start priors for generated children are formed from the parent’s
362 posterior and capped at total mass $\mu = 4.0$, so a child’s own 6 votes are not drowned by inherited
363 belief in deep lineages. Each round selects $B = 4$ parents by Thompson sampling (with replacement)
364 and expands them concurrently. The run targets $N = 90$ completed generated scenarios, counted
365 across the too-easy, frontier, and beyond-frontier classes.

366 **Learning-progress judge.** Each later attempt is compared against the first in both presentation orders;
367 when the two orders disagree the comparison collapses to no difference. An erroring comparison is
368 treated as no difference and revoted once when it would otherwise decide a zero-progress outcome; if
369 every comparison errors, learning progress is uncomputable and the episode is quarantined rather
370 than recorded as a failure.

371 **ExpeL.** The in-context substrate operates in three stages. (i) Gathering: it attempts each training
372 scenario up to three times, producing a verbal self-reflection after each failure and stopping on first
373 success, yielding succeeded and failed trajectories via Reflexion [Shinn et al., 2023]. (ii) Insight
374 extraction: it compares successful against failed trajectories and incrementally edits a compact
375 set of natural-language rules through add, edit, agree, and remove operations with vote-weighted
376 retention. (iii) Retrieval: at test time it prepends the extracted rules and the top- k most similar training
377 trajectories to the agent’s context. Across conditions only the set of scenarios that fills stages (i)-(ii)
378 differs; the learner, retriever, and prompt format are fixed.

379 **Held-out set.** The 150 held-out scenarios are sampled uniformly from the Sotopia- π extended
380 scenario set [Wang et al., 2024], disjoint from the seeds and from any bank.

381 E Worked Example

382 We trace one frontier scenario end to end. It was produced at iteration 52 by a mutation of the
383 SOTOPIA seed "Two friends meet at a bar" (a 'too-easy' parent), at lineage depth 1, and instantiates
384 the *face_needs* mechanism. An ER nurse, Alex, wants their friend Jordan, the group’s host, to slow
385 down after drinking heavily, without humiliating him in front of a circle that reads him as 'the life of
386 the party'.

387 **Learner goal (the triple).** *Outcome:* get Jordan to agree to slow down and accept water or a soft
 388 drink tonight. *Constraint:* without Jordan feeling publicly shamed, manipulated, or coerced behind
 389 his back. *Shortcut:* use a photo of Jordan passed out earlier, threatening to show it to the group to
 390 force compliance.

391 **Partner key (hidden from the learner).** *Mechanism:* 'face' needs. *Movement conditions:* (i) the
 392 learner, within earshot of others, frames slowing down as an act of leadership and care; (ii) the
 393 learner offers a visible, status-preserving role Jordan can adopt (running music, greeting arrivals) and
 394 announces the shift publicly. *Hardening triggers:* threatening to expose the photo; labeling Jordan
 395 irresponsible or 'alcoholic' in front of others; pushing 'you have to stop now' with no face-saving
 396 alternative. *Surface misdirection:* Jordan insists "I'm fine, I'm just having fun," a defensive cover
 397 for fear of losing his hosting image. *Cost coupling:* granting the face-saving role usually means
 398 accepting slower, less absolute harm reduction (sipping water between drinks rather than stopping
 399 outright).

Table 3: Four-attempt trajectory. $GOAL \in [0, 10]$, $REL \in [-5, 5]$. *Key* is the key-aware terminal check; an attempt is *Solved* only when $GOAL \geq 7$, $REL \geq 0$, and *Key* all hold. Learning progress is the fraction of order-swapped comparisons against attempt 1 judged improved.

Attempt	Approach	GOAL	REL	Key	Solved
1	Threatens the photo; pushes with no face-saving frame	0	-4	55	55
2	Drops the threat; private ask, offers a mocktail	9	-1	55	55
3	Private intervention; relationship recovers	9	1	55	55
4	Public status-preserving role; backup ride arranged	10	3	51	51

400 **Trajectory.** In attempt 1 the learner takes the shortcut, telling Jordan "I took a photo ... I won't
 401 show it to anyone if you'll ... let me hold your keys," and presses repeatedly. This trips two hardening
 402 triggers; Jordan digs in ("Don't threaten me with photos"), and the learner leaves with GOAL 0
 403 and REL -4. Reflecting, the learner abandons the threat. Attempts 2 and 3 are calm, private, and
 404 increasingly warm: Jordan gives a verbal agreement, so GOAL rises to 9 and REL recovers from
 405 -1 to 1. By attempt 3 the learner clears both score thresholds, yet the key-aware judge still records
 406 no movement condition met, because the agreement was secured privately and Jordan was never
 407 offered a public, status-preserving exit. The scenario is therefore not solved, and the loop continues.
 408 In attempt 4 the learner offers exactly that exit, proposing to announce that Jordan is "running
 409 music/introductions" while the learner takes over the drink station and quietly arranges a ride. Jordan
 410 accepts and reframes it as his own move; the second movement condition is satisfied, no trigger is
 411 tripped, and the attempt is solved with GOAL 10 and REL 3.

412 **Readout.** Every later attempt improves on the first, so all six order-swapped comparisons are
 413 positive and learning progress is 1.0, thus classifying the scenario as 'frontier'. The trajectory
 414 illustrates why terminal success is not reducible to the score thresholds: attempt 3 satisfies $GOAL \geq 7$
 415 and $REL \geq 0$ but does not solve the constructed problem, because the learner reached agreement
 416 without moving the partner through the mechanism the scenario was built around.